Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

CHEM 452

IONS IN SOLUTION

Lecture 1-3

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COURSE OUTLINE

- 1. Kinetics and Mechanisms of reactions of Complex ions.

  (A) Stability of complexes in aqueous solution:

- Definition of stability; Thermodynamic stability and Kinetic stability
- Lability and inertness
- Factors affecting stability of complexes
- HSAB (class a and class b metals)
- Experimental determination of stability constant and composition of a complex

(B) Ligand substitution reactions in octahedral complexes.

- Transition state or activated complexes
- Types of substitution reactions
- Kinetics of octahedral substitution reactions

COURSE OUTLINE

(C)Substitution reactions in Square-planar Complexes.

- The trans-effect
- Theories of the trans-effect
- Mechanism of substitution reactions

(D) Electron-Transfer (Redox reactions) in coordination compounds.

- One Electron-Transfer reactions
- Direct Electron-transfer or outer-sphere mechanism
- Indirect Electron-transfer or inner-sphere mechanism
- Two electron-transfer reactions
- Complementary and non-complementary electron-transfer reactions of Coordinated Ligands
- Reactions of coordinated ligands

REFERENCES

- Inorganic Chemistry by Duward F. Shriver, Peter W. Atkins and Cooper H. Langford.
- Concepts and Models of Inorganic Chemistry by Bodie E. Douglas, Darl H. McDaniel and John J. Alexander.
- Inorganic Chemistry by J.E. Huheey et al.
- Advanced Inorganic Chemistry by F. A. Cotton, Geoffrey Wilkinson et al.
- Inorganic Chemistry by Gary L. Miessler and Donald A. Tarr, Pearson Education International Prentice Hall.

INTRODUCTION

Here are some suggestions to help you master the course;

- ✓ *Attend classes regularly* and take careful notes.
- ✓ Always *review topics discussed* in class the same day they are covered.
- ✓ *Think critically* and ask intelligent questions in class.
- ✓ Do not hesitate to ask your instructor or your TA for help.

COURSE EVALUATUON

- You will start this course with the assumed grade of an A.
- Your responsibility will be to work to keep this grade.
- You will fulfill this responsibility by meeting the following requirements:
- Attendance, Exercises, Exams and Class Contributions.

Attendance Policy

- There will be regular roll call during classes.
- There will be no unexcused absence.
- Your final grade will be drop for each unexcused absence.
- This adjustment to your final grade will be made after your regular grade has been determined.

I reserve the right to change any part of these should it become necessary.

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OBJECTIVES

- The subject of the mechanistic study of ligand substitution reactions is currently undergoing an exciting growth. New fast-reaction techniques have removed the upper limit on rates that can be measured, and extension to less familiar central metal atoms has begun in earnest.
- This might seem the wrong moment for review of the field. As yet, definitive treatment is possible only for those cornplexes involving monodentate ligands with cobalt(II1) and platinurn(I1).
- But, because information is so extensive for these systems, it is clear that they are functioning as models from which concepts and experiments are generated for application over the fast-growing range of the subject.

OBJECTIVES

- This course module therefore seeks to deal with the reactions of coordination compounds which share some characteristics with reactions of other molecules, both organic and inorganic, so an understanding of coordination compound reactions can draw on some familiar concepts.
- However, the chemistry of coordination compounds has some additional features because the molecules have more complex geometries and more possibilities for rearrangement, the metal atoms exhibit more variability in their reactions, and different factors influence the course of reactions.

OBJECTIVES

- **Reactions of coordination complexes can be conveniently divided into:**
  - Substitution reactions at the metal center
  - Oxidation-reduction reactions
  - Reactions of the ligands that do not change the attachments to the metal center
- Synthesis of coordination compounds has always been a major part of chemistry.
- Although the early chemists did not know the structures of the compounds they worked with, they did learn how to make many of them and described them according to the style of the time.
- Many years of experimentation and consideration of possible reaction pathways have led to the ideas described in this course, and even now these ideas must be considered tentative and provisional in many cases.
- Although the ability to predict products and choose appropriate reaction conditions to obtain the desired products is still a matter of art as well as science, the list of known reactions is now long enough to provide considerable guidance.

OBJECTIVES

- The goals of those studying reaction kinetics and mechanisms vary, but a major underlying reason for such studies is to understand the electronic structure of the compounds and their interactions.

- The information from these studies also allows more control of reactions and the design of reaction steps that may be useful for **synthesis.**

- A by-product of synthetic and kinetic studies is the esthetic pleasure of seeing the colours that are characteristic of many coordination compounds and how they change with changes in ligands and metal ions.

Stability of Complexes in Aqueous Solutions

- **Definition:** The term "**stability**' can be used in a number of different ways. The statement that a complex is 'stable' is rather loose since several interpretations can be placed on it.

- Using the term "stability" without qualification, means that the complex exists and under suitable conditions, it may be stored for a long time.

NB: This term cannot be generalized for complexes, since a complex may be quite stable to one reagent and may decompose readily in the presence of another reagent. This term may also refer to the action of heat or light on a compound.

Stability of Complexes in Aqueous Solutions

- In studying the formation of complexes in solution, two kinds of stability of complexes are considered:

I. Thermodynamic Stability: This is a measure of **the extent** to which the complex will form or will be transformed into another species (under certain conditions), when the system has reached equilibrium. This type of stability deals with metal-ligand bond energies, stability constants, etc.

II. Kinetic Stability: This refers to the **speed** with which transformations leading to the attainment of equilibrium will occur. When we are concerned with kinetic stability of primarily complex ions in solutions, we deal with rates and mechanism of chemical reactions such as, substitution, isomerization, racemization and electron or group transfer reactions, as well as with the thermodynamic variables involved in the formation of intermediate species or activated complexes.

Stability of Complexes in Aqueous Solutions

- The quantity related to kinetics is the rate constant, **K**; this constant is associated with the activation energy required for the reaction to proceed, that is, the reactivity of the reactants.

$$k = Ae^{-\frac{E_a}{RT}} \qquad ln(k) = -\frac{E_a}{RT} + lnA$$ Arrhenius equation

- The thermodynamic quantity is the energy difference resulting from the free energy (ΔG) given off during a chemical reaction; the stability of the products relative to the reactants.
- Although kinetics describes the rates of reactions and how fast equilibrium is reached, it gives no information about conditions once the reaction equilibrates.
- In the same measure, thermodynamics only gives information regarding the equilibrium conditions of products after the reaction takes place, but does not explain the rate of reaction.

Stability of Complexes in Aqueous Solutions

Thermodynamics and Stability

$\Delta G^o = -RT \ln k$

$\Delta G^o = \Delta H^o - T\Delta S^o$

- The lower the potential energy of the system, the more stable it is.
- Chemical processes usually occur because they are thermodynamically favourable (*i.e.* $\Delta \mathbf{G}$ = -ve).

"**Thermodynamically favourable**" means from high energy to low energy, or, put another way, from less stable to more stable.

- Understanding the relative stability of molecules can be important for predicting relative reactivity of starting materials and the relative yields of potential products.
- Stability can be determined by comparing experimentally measured or based on theoretical calculations.

Inert and Unstable Complex

- $[Co(NH_3)_6]^{3+}$ reacts slowly.
- When this complex is treated with concentrated HCl, no reaction takes place.
- Only when it is heated with 6M HCl for many hours, one NH_3 is substituted by Cl^-.

$$[Co(NH_3)_6]^{3+} + HCl \rightarrow [Co(NH_3)_5Cl]^{2+} + NH_4^+$$

Stability of Complexes in Aqueous Solutions

- In the kinetic sense it is more appropriate to refer to complexes as **inert or labile** complexes rather than stable or unstable complexes.

NB: The complexes in which the ligands are readily replaced by others are referred to as LABILE while those in which substitution occurs slowly are INERT complexes. This depends on the stepwise formation of complexes; stepwise formation constants and overall formation constants.

- The formation of the complex MLn (M = central metal cation, L= monodentate ligand and n= maximum coordination number of the metal ion M for the ligand L. n varies from one ligand to another for the same metal ion) may be supposed to take place by the following n consecutive steps and equilibrium constants:

Stability of Complexes in Aqueous Solutions

- Assuming that no insoluble products are formed, nor species containing more than one metal. The interaction of M and L may be formulated as a series of successive equilibra:

$M + L \rightleftharpoons ML$	$K_1 = \frac{[ML]}{[M][L]}$
$ML + L \rightleftharpoons ML_2$	$K_2 = \frac{[ML_2]}{[ML][L]}$
$ML_2 + L \rightleftharpoons ML_3$	$K_3 = \frac{[ML_3]}{[ML_2][L]}$

$ML_{(n-1)} + L \rightleftharpoons ML_n$	$K_n = \frac{[ML_n]}{[ML_{(n-1)}][L]}$

Stability of Complexes in Aqueous Solutions

$M + L \rightleftharpoons ML$	$K_1 = \frac{[ML]}{[M][L]}$
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$ML_{(n-1)} + L \rightleftharpoons ML_n$	$K_n = \frac{[ML_n]}{[ML_{(n-1)}][L]}$

- Where n is the maximum number of ligands attached to one central metal for a particular set of experimental conditions and, therefore represents its coordination number. The equilibrium constants K_1, K_2, $K_3 \ldots K_n$ are known as the STEPWISE FORMATION OR STABILITY CONSTANT.

Stability of Complexes in Aqueous Solutions

- Another way of expressing the equilibrium relation is the following, for which the general symbol β (overall stability constant) is used:

$M + L \rightleftharpoons ML$	$\beta_1 = \frac{[ML]}{[M][L]}$
$M + 2L \rightleftharpoons ML_2$	$\beta_2 = \frac{[ML_2]}{[M][L]^2}$
$M + 3L \rightleftharpoons ML_3$	$\beta_3 = \frac{[ML_3]}{[M][L]^3}$

$M + nL \rightleftharpoons ML_n$	$\beta_n = \frac{[ML_n]}{[M][L]^n}$

Stability of Complexes in Aqueous Solutions

- Since there can be only n independent equilibra in such a system, it is clear that the K's and β's must be related by the equation below.
- Consider β_3 expression and multiplying the numerator and the denominator by $[M][ML_2]$ and rearranging the expression gives:

$$\beta_3 = \frac{[ML_3]}{[M][L]^3} \times \frac{[ML][ML_2]}{[ML][ML_2]}$$

$$= \frac{[ML]}{[M][L]} \times \frac{[ML_2]}{[ML][L]} \times \frac{[ML_3]}{[ML_2][L]}$$

$$\beta_3 = K_1 \times K_2 \times K_3$$

$$\beta_n = K_1 \times K_2 \times K_3 \ldots\ldots K_n$$

OVERALL →

$$\beta_n = \prod_{i=1}^{i=n} K_n$$

Stability of Complexes in Aqueous Solutions

- ✓ The K_i's are called the STEPWISE FORMATION CONSTANTS (or stepwise constants), and the β_i's are called the OVERALL FORMATION CONSTANTS (or overall stability constants); each type has its special convenience in certain cases.
- ✓ βn is the nth overall formation or stability constant.
- ✓ The higher the value of stability constant for a complex ion, the greater will be its stability.
- ✓ Alternatively 1/K values sometimes called instability/dissociation constant, gives a measure of the extent to which the equilibrium representing the formation of a complex lies to the right.
- ✓ It is important to note that ML to ML_n may exist in the system but their proportions may depend on the relative concentrations of M and L and their K_i values.
- ✓ With only a few exceptions, there is generally slowly descending progression in the values of the K_i's in any particular system. For example in Cd^{II}-NH_3.

Stability of Complexes in Aqueous Solutions

$Cd^{2+} + NH_3 \rightleftharpoons [Cd(NH_3)]^{2+}$ $K = 10^{2.65}$

$[Cd(NH_3)]^{2+} + NH_3 \rightleftharpoons [Cd(NH_3)_2]^{2+}$ $K = 10^{2.10}$

$[Cd(NH_3)_2]^{2+} + NH_3 \rightleftharpoons [Cd(NH_3)_3]^{2+}$ $K = 10^{1.44}$

$[Cd(NH_3)_3]^{2+} + NH_3 \rightleftharpoons [Cd(NH_3)_4]^{2+}$ $K = 10^{0.93}$

Exercise

Calculate the overall complex dissociation equilibrium constant for the $Cu(NH_3)_4^{2+}$ ion, given that β_4 for this complex is 2.1×10^{13}.

ANS: The overall dissociation constant is the reciprocal of overall stability constant i.e. $1/\beta_4 = 4.7 \times 10^{-14}$

Stability of Complexes in Aqueous Solutions

- ✓ If the values of stepwise stability constants for a series of successive reactions are logK1 = 4, logK2 = 3.2, logK3 = 2.7 and logK4 = 2, what is the value of β4?

ANS

- ➢ Clarification: logβ4 = logK1 + logK2 + logK3 + logK4
- ➢ log β4 = 4 + 3.2 + 2.7 + 2 = 11.9
- ➢ β4 = antilog(11.9)
- ➢ $\beta 4 = 7.94 \times 10^{11}$

REASONS FOR THE REDUCTION IN K

These are:

1. Statistical factors
2. Stearic factors
3. Columbic factors

NB: If the incoming ligand can change the oxidation state of the metal i.e. increases the oxidation state of the metal, stronger bond is formed.

Statistical Reasons

- When more ligands are entering into the coordination sphere the number of aqua ligand decreases.
- This reduces the probability of substitution of aqua ligand with the new ligand.
- Reflected as decreasing stepwise formation constants.

REASONS FOR THE REDUCTION IN K

- Additionally, ML_2 depends on ML. ML_3 depends on ML_2 i.e. subsequent reactions depend on the previous amount present.
- Thus further reaction will be difficult due to the small available space around the central metal. Thus further reaction will be difficult due to the small available space around the central metal ion. This reduces the speed of reaction and K_n values.
- In general, the steady decrease in the values of K_1 , K_2 , ….. K_n with increasing number of ligands is due to the fact that as more and more ligands move into the coordination zone, less and less aqua-molecules are available to fresh ligands for replacement.

Stearic factors

- When the ligand is bulkier than water, with the displacement of the water molecule by the ligand, it reduces the space around the central metal.

REASONS FOR THE REDUCTION IN K

Columbic factors

- With progressive intake of ligands, the metal ion becomes less electron greedy. i.e., $K_1 > K_2 > K_3 \ldots \ldots \ldots \ldots \ldots \ldots > K_{n-1} > K_n$.

- This also has got to do with charged ligands. The negative charges on the ligands could cause repulsion as a result, it reduces the rate of reaction, hence K_n values.

Kinetic vrs thermodynamic Stability

- Since the terms; labile and inert show the speed or rates at which the substitution of one ligand by other occurs, these terms represent the kinetic stability of complexes.

- These terms should, therefore, not be confused with or used for thermodynamic stability terms viz, unstable and stable respectively.

- Although thermodynamically stable complexes may be labile or inert, and unstable complexes which are usually labile may also be inert, there is no correlation between thermodynamic and kinetic stability terms.

Kinetic vrs Thermodynamic Stability

- For example $[Hg(CN)_4]^{2-}$ which is thermodynamically very stable (formation constant, 10^{42}) is labile, since in solution it exchanges CN ligands with labelled cyanide ions, $^{14}CN^-$ at a very fast rate.

$$[Hg(CN)_4]^{2-} + {}_4{}^{14}CN^- \underset{\text{Fast}}{\rightleftharpoons} [Hg(^{12}CN)_4]^{2-} + {}_4CN^-$$

- Thus the stability of $[Hg(CN)_4]^{2-}$ does not ensure its inertness. On the other hand the complex $[Co(NH_3)_6]^{3+}$ which is thermodynamically unstable can remain unchanged in acid solution for weeks. Thus this complex is unstable but inert in acid solution.

Conclusion

- Complexes in which exchange of one or more ligands are rapidly exchanged are called labile complexes.
- If the rate of ligand exchange is slow then the complex is said to be inert.
- Lability is not related to the thermodynamic stability of a complex.
- A stable complex may be labile or inert , so as the unstable complex.
- Inert complexes are not necessarily thermodynamically unstable and that labile complexes are not necessarily thermodynamically unstable.
- The stability of a complex depends on the reaction energy while the lability of a compound depends on the activation energy.
- The larger the activation energy, the lower will be the lability of a compound, i,e., the compound with larger activation energy will react slowly.

Labile and Inert Octahedral Complexes-VBT

Labile and Inert Octahedral Complexes according to the VBT

- With the help of the VBT it is possible to predict which octahedral complex is labile and which is inert.

- VBT classifies octahedral complexes into two types.

➢ Inner orbital complexes – d^2sp^3

➢ Outer orbital complex – sp^3d^2

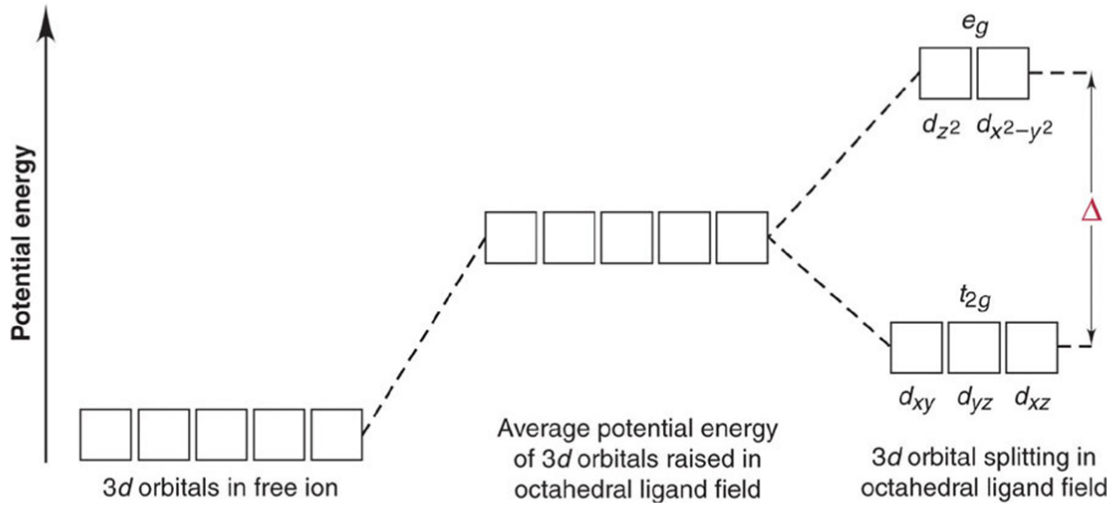
➢ The two d-orbitals involved in the hybridization are the e_g set of orbitals ie. dx^2-y^2 and dz^2 orbitals (e_g set).

Labile and Inert Octahedral Complexes-VBT

- It is assumed that d^2sp^3 orbitals from stronger bonds, providing the energy required to cause electron pairing. Since it contains no unpaired electrons, this complex is diamagnetic in contrast to knowledge of the magnetic properties of a complex.
- However, it is also not always possible to make the reverse prediction. For example, it may be predicted that octahedral complex of iron (III) would have a magnetic moment corresponding to a single unpaired electron and that octahedral complexes of $[FeF_6]^{3-}$ and $[CoF_6]^{3-}$ have five and four unpaired electron respectively.
- To account for their octahedral geometries, the following configuration must be used.

Labile and Inert Octahedral Complexes-CFT

Splitting of *d*-orbital energies in an octahedral field of ligands



metal ion in octahedral complex

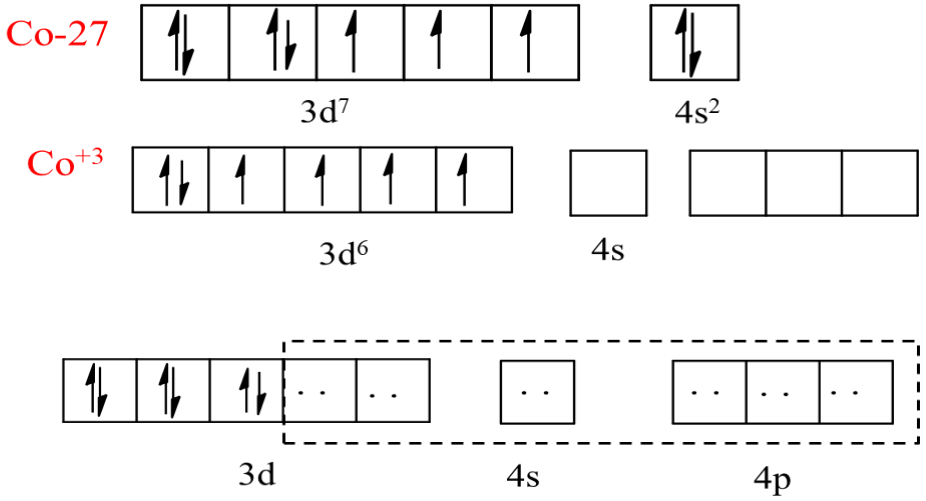
The *d* orbitals split into two groups. The difference in energy between these groups is called the ***crystal field splitting energy***, symbol Δ.

$$CFSE_{Oct.} = -0.4n(t_{2g}) + 0.6n(eg)$$

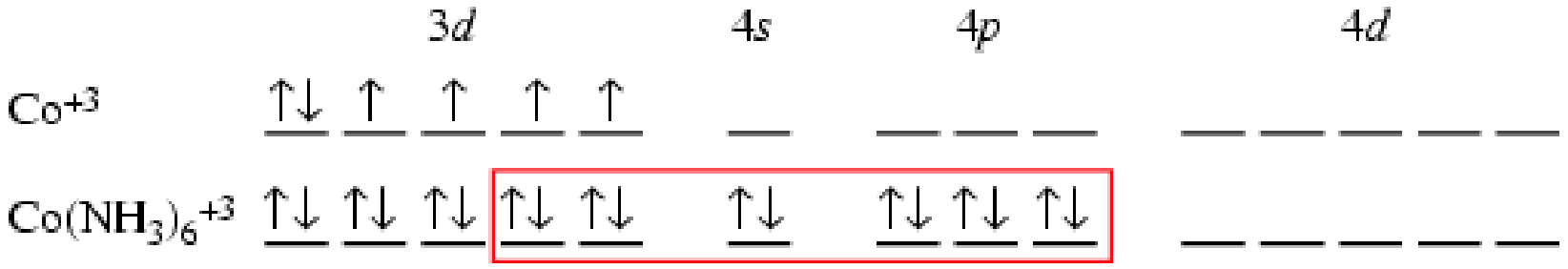
Labile and Inert Octahedral Complexes-VBT

Hexaamminecobalt (III) ion; $[Co(HN_3)_6]^{+3}$
Or $[Co(CN)_6]^{-3}$

Inner orbital complexes are stronger and inert

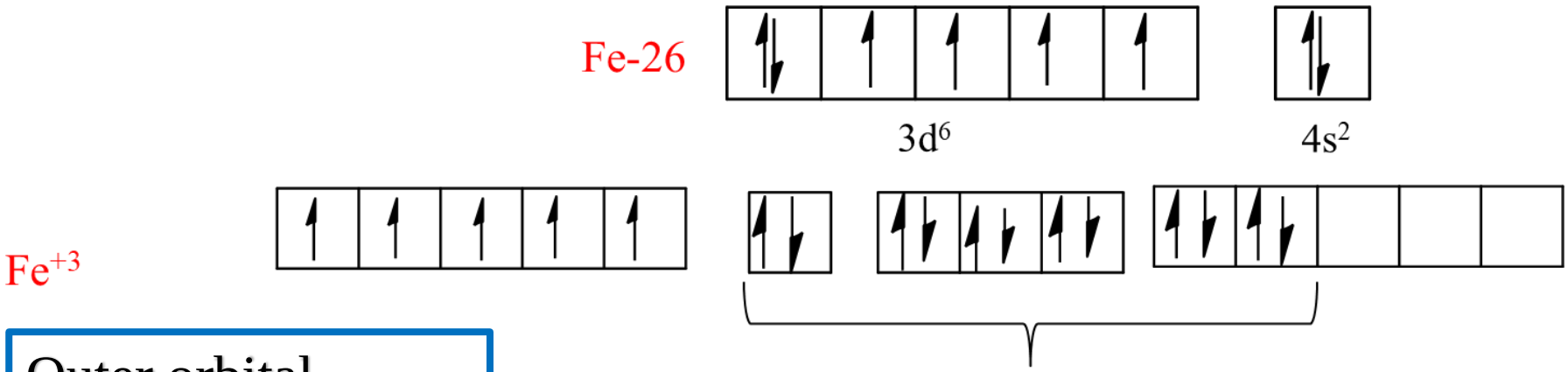


To form octahedral complex of Co(III) using d^2sp^3 hybrid orbitals there must be a rearrangement of the electrons so that two empty d orbitals are provided.



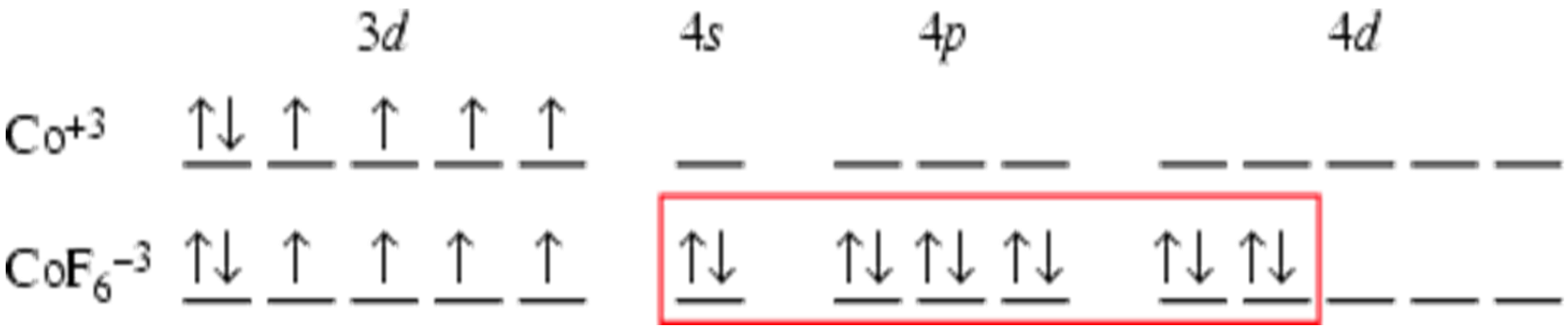
Labile and Inert Octahedral Complexes-VBT

Hexafloroferrate (II) ion; $[FeF_6]^{3-}$



Outer orbital complexes are weaker and labile

sp^3d^2 hybridization-diamagnetic -octahedral



Labile and Inert Octahedral Complexes-VBT

- **Outer orbital complexes** have magnetic moments as high as that of free metal ion, such complexes are called high spin or spin free.

- Inner orbital complexes usually have fewer unpaired electrons that the free metal ion and they are termed low spin or spin paired.

Labile and Inert Octahedral Complexes-VBT

(i) Outer-orbital octahedral complexes

- In terms of VBT these bonds are weaker and labile. e.g. the octahedral complexes of Mn^{2+} ($3d^5$), Fe^{2+} ($3d^6$), Fe^{3+} ($3d^5$), Co^{2+} ($3d^7$), Ni^{2+} ($3d^8$), Cu^{2+} ($3d^9$) and Cr^{2+} ($3d^4$) exchange ligands rapidly and hence are labile. In terms of VBT, this observation can be correlated with the weakness of the bonds of sp^3d^2 type as compared with d^2sp^3 bonds.

(ii) Inner-orbital Octahedral Complexes

- These complexes generally have d^2sp^3 hybridization.
- The hybrid orbitals are filled with the ligand electrons.
- The t_{2g} orbitals of metal forms the d electrons of the metal
- Since the six d^2sp^3 hybrid orbitals are filled with the six electrons pairs donated by the six ligands, d^n electrons of the central metal will occupy dxy, dyz and dxz orbitals.

Labile and Inert Octahedral Complexes

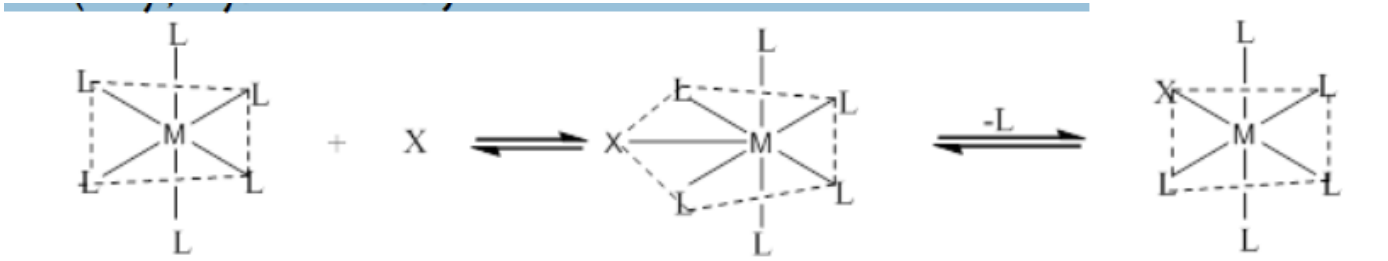
Table: Distribution of d^n electrons in the various t_{2g} orbitals for labile and inert inner-orbital octahedral complexes according to VBT

Type of the complex	d^n configuration	Distribution of d^n electrons (shown by arrows) in t_{2g} orbitals. Electrons shown by crosses in the eg orbitals have been donated by six ligands to enter d^2sp^3 hybrid and are in opposite spins									Examples of central metal ions
		d					s	p			
		t_{2g}			e_g						
		xy	yz	zx	X^2-y^2	Z^2		p_x	P_y	P_z	
Inner-orbital labile octahedral complexes	d^0				X X	X X	X X	X X	X X	X X	Sc(+3), Y(+3), rare earth (+3), Te(+3), Zr(+4), Hf(+4), etc.
	d^1	↑			X X	X X	X X	X X	X X	X X	Ti(+3), V(+4), Mo (+5), W(+5), Re(+6)
	d^2	↑	↑		X X	X X	X X	X X	X X	X X	Ti(+2), v(+3), Nb(+3), Ta(+3), W(+4), Re(+5), Ru(+6),
Inner-orbital Inert octahedral complexes	d^3	↑	↑	↑	X X	X X	X X	X X	X X	X X	V(+2), Cr(+3), Mo (+3), W(+3), Mn(+4), Re(+4)
	d^4	↑↓	↑	↑	X X	X X	X X	X X	X X	X X	$[Cr(CN)_6]^{4-}$, $[Mn(CN)_6]^{3-}$, Re(+3), Ru (+4), Os(+4),
	d^5	↑↓	↑↓	↑	X X	X X	X X	X X	X X	X X	$[Mn(CN)_6]^{4-}$,Re(+2), Ir(+4) $[Fe(CN)_6]^{3-}$, Ru(+3), Os(+3),
	d^6	↑↓	↑↓	↑↓	X X	X X	X X	X X	X X	X X	$[Fe(CN)_6]^{4-}$, Ru(+2), Os(+2), Co(+3) (except $CoFe_6^{3-}$), Rh(+3), Ir(+3)

Labile and Inert Octahedral Complexes

POINTS TO NOTE

- In the labile and inner-orbital octahedral complexes, there is at least one d-orbital of t_{2g} set empty so that this empty d-orbital may be used to accept the electron pair from the incoming ligand in forming the transition state (unstable intermediate) with coordination number of seven. This leads to extremely fast reactions of these complexes. It has not been possible to determine whether the intermediate exists in these reactions.
- In the inert inner-orbital octahedral complexes every d-orbital of t_{2g} set (dxy, dyz and dxz) contains at least one electron.



Labile inner-orbital octahedral complex ML_6 with one-empty d-orbital of t_{2g} set

Seven-coordinated unstable intermediate ML_6X (CN=7)

Octahedral complex ML_5X (CN=6)

Summary

Outer orbital complexes

- The complexes having sp^3d^2 hybridization are called outer orbital complexes.
- In terms of VBT these bonds are weaker.
- They are generally labile.
- Mn(II), Fe(II),Fe(III),Co(II),Ni(II),Cu(II) and Cr(II) are labile.

Inner orbital complexes

- These complexes generally have d^2sp^3 hybridization.
- The hybrid orbitals are filled with the ligand electrons.
- The t_{2g} orbitals of metal accommodate the d electrons of the metal.

If the t_{2g} levels are left vacant then the complex can associate with an incoming ligand and the complex is labile

If all the t2g levels are occupied then the complex becomes inert.

Labile and Inert Octahedral Complexes-CFT

- ✓ According to CFT the ligand field splits the d -orbitals.
- ✓ This splitting leads to a decrease in energy of the system whose magnitude depends on the number of d electrons present.
- ✓ If the CFSE value increases by association or dissociation of a ligand then the complex is labile.
- ✓ On the other hand it is inert when there is a loss in CFSE value.
- ✓ Whether an octahedral complex reacts by an S_N^1 dissociation or S_N^2 association mechanism, the symmetry is lowered and a decrease in CFSE usually occurs in going from octahedral to 5-coordinate square pyramidal activated intermediate or from octahedral to 7-coordinate pentagonal bipyramidal transition state.
- ✓ The change in CFSE in going from octahedral to square pyramidal (S_N^1 mechanism) and in going from octahedral to pentagonal bipyramidal (S_N^2 mechanism) structure is called **activation energy** and is denoted by Ea.

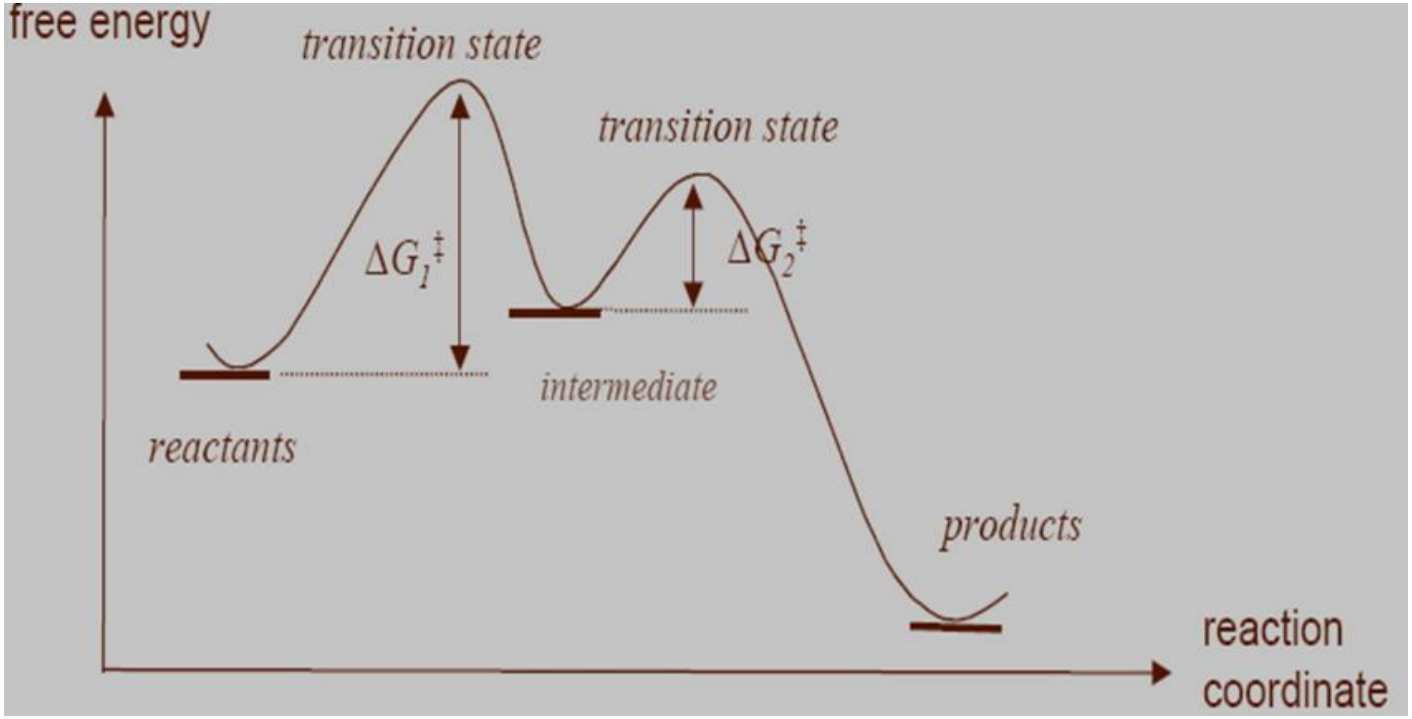
Labile and Inert Octahedral Complexes-CFT

- ✓ It has been assumed that there is no loss in CFSE when they are changed into activated complexes.
- ✓ The octahedral complexes formed by the ions for which there is large loss in CFSE are least labile, i.e. such complexes are INERT.
- ✓ On the other hand octahedral complexes given by ions for which there is little or no loss in CFSE are labile, i,e. such complexes react rapidly.

Labile and Inert Octahedral Complexes-CFT

- ✓ We know that CFT assumes the splitting of d-orbitals of metal.
- ✓ Filling of e-s in them results in different CFSE.
- ✓ CFAE – Crystal Field Activation Energy

  CFAE = CFSE of intermediate – CFSE of Reactant.
- ✓ Since the geometries of the reactant and intermediate are different their splitting and CFSE are also different.



Conclusions Based on CFAE

- ✓ If the calculated CFAE is negative or zero or low the reacting complex will require less energy to form the intermediate, hence it will be labile. By oversimplified approach the –ve values of CFAE may be taken as zero.
- ✓ If CFAE is a high positive value then the complex will be inert.
- ✓ It must be borne in mind that CFAE is only a part of actual AE and other factors are also operative.

FACTORS AFFECTING STABILITY OF THE COMPLEXES

- Properties of the metal ion

✓ Radii of the central ion

✓ Charge and size of the ion

✓ Effect of ionic radius

✓ Effect of charge

✓ Class a and Class b metals

✓ Electronegativity of the metal atom

✓ Natural order (or) Irving –William order of stability.

✓ d-electron configuration

FACTORS AFFECTING STABILITY OF THE COMPLEXES

- Properties of ligand

✓ Size and charge of the ligand

✓ Basic character

✓ Chelate effect

✓ Size of the chelate ring

✓ Steric effect

Properties of the Metal Ion

✓ Charge and size.

- In general metal ions with higher charge and small size form stable complexes.
- A small cation with high charge attracts the ligands more closely leading to stable complexes.
- The following tables explain the facts that if z/r ratio (polarizing power) of the metal ion is high then stability of the complex is also high.

Properties of the Metal Ion

✓ Effect of ionic radius

Complex ion	Charge on the ion	Ionic radii (A°)	Value of β	stability
$[Be^{II}(OH)]^{+}$	+2	0.31	10^7	
$[Mg^{II}(OH)]^{+}$	+2	0.65	120	
$[Ca^{II}(OH)]^{+}$	+2	0.99	30	
$[Ba^{II}(OH)]^{+}$	+2	1.35	4	



Properties of the Metal Ion

✓ Effect of charge

Complex ion	Charge on the ion	Ionic radii (A°)	Value of log β	stability
$[Fe^{III}(CN)_6]^{3-}$	+3	Almost same	31.0	Stability increases
$[Fe^{III}(CN)_6]^{4-}$	+2		8.3	
Co^{III} complex	+3	Almost same	high	Stability increases
Co^{II} complex	+2		low	

Properties of the Metal Ion

✓ Irving – William order of stability

- Because of the strong electrostatic field around the cation, the smaller the size and the larger the charge of the metal ion the more stable are the metal complexes.
- The order of stability of the complexes formed by the stable positive ions of the first transition series (from manganese onwards, irrespective of the ligands, is first related to the ionic radius:
- **Order of stability:** $Mn^{2+} < Fe^{2+} < Ni^{2+} < Cu^{2+} < Zn^{2+}$ ionic radius (M^{2+}/A^{o}) 0.91, 0.83, 0.82, 0.78, 0.69, 0.74 respectively.
- This order is called the **natural order of stability** and is consistent with the charge- to –radius concept.

• This is attributed to the CFSE values of the complexes.

Properties of the Metal Ion

✓ Class a and Class b metals

Chatt and Ahrland classified metals into three types:

- **Class a, Class b and border line**.
- **Class a** : H, alkali and alkaline earth metals, Sc -> Cr, Al -> Cl, Zn -> Br , In, Sn , Sb , I, lathanides and actinides
- **Class b:** Rh ,Pd , Ag , Ir , Pt , Au and Hg
- The class b acceptors form the most stable complexes with ligand atoms from the third or later period.
- Border line: Mn -> Cu , Tl -> Po, Mo , Te , Ru , W , Re , Os and Cd;

Pearson extended the concept into a broad range of acid-base interactions.

Properties of the Metal Ion

- Class a metals are **small** and not very polarizable and these prefer ligands that are small and not very polarizable.
- Pearson referred to these metals as HARD ACIDS and the ligands as HARD BASES.
- The class b metals are **larger** and more polarizable and prefer larger and more polarizable ligands.
- These metals are referred to as SOFT ACIDS and the ligands, SOFT BASES.
- The relationship is that hard acids prefer to react with hard bases and soft acids react with soft bases.

Properties of the Metal Ion

- ✓ Class a metals form more stable complexes with ligands in which coordination atoms are from second period. (N, O, F) eg. (ammonia, amines, water, ketones, alcohols) etc.
- ✓ Class b metals form more stable complexes with ligands having third period elements as ligating atoms (P , S, Cl).
- ✓ Class b metals are having capacity to form pi bonds with the ligand atoms. The expansion is possible only from the third period donor atoms.
- ✓ Border line metals do not show any noticeable trend.

Pearson Classification of hard and soft behavior

Hard Acids	Borderline Acids	Soft Acids
H^+, Li^+, Na^+, K^+		
Be^{2+}, Mg^{2+}, Ca^{2+}, Sr^2		
BF_3, BCl_3, $B(OR)_3$	$B(CH_3)_3$	BH_3, Tl^+, $Tl(CH_3)_3$
Al^{3+}, $Al(CH_3)_3$, $AlCl_3$, AlH_3		
Cr^{3+}, Mn^{2+}, Fe^{3+}, Co^{3+}	Fe^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Zn^{2+} Rh^{3+}, Ir^{3+}, Ru^{3+}, Os^{2+}	Cu^+, Ag^+, Au^+, Cd^{2+}, $Hg_2{}^{2+}$, Hg^{2+}, CH_3Hg^+, $[Co(CN)_5]^{3-}$, Pd^{2+}, Pt^{2+}, Pt^{4+},
Ions with formal oxidation states of 4 or higher		Br_2, I_2
HX (hydrogen-bonding molecules)		Metals with zero oxidation state π acceptors: e.g.,

Pearson Classification of hard and soft behavior

Hard Bases	Borderline Bases	Soft Bases
		H^-
F^-, Cl^-	Br^-	I^-
H_2O, OH^-, O^{2-}		H_2S, SH^-, S^{2-}
ROH, RO^-, R_2O, CH_3COO^-		RSH, RS^-, R_2S
NO_3^-, ClO_4^-	NO_2^-, N_3^-	SCN^-, CN^-, RNC, CO
CO_3^{2-}, SO_4^{2-}, PO_4^{3-}	SO_3^{2-}	$S_2O_3^{2-}$
NH_3, RNH_2, N_2H_4	$C_6H_5NH_2$, C_5H_5N, N_2	PR_3, $P(OR)_3$, AsR_3, C_2H_4, C_6H_6

Properties of the Metal Ion

✓Electronegativity of the metal atom

- ❖The bond between metal and ligand atom is, to some extent due to the donation of electron pair to the metal.
- ❖If the metal is having a tendency to attract the electron pair (higher electronegativity) then more stable complexes are formed.

Factors Affecting Lability of Complexes

✓ d-electron configuration:

- If electrons are present in the antibonding e_g^* orbitals, the complex will be labile because the ligands will be weakly bonded to the metal and hence can be substituted easily.
- If the complex contains empty t_{2g} orbitals, then it will be labile because ligands can approach easily without much repulsion.
- In short, if the complex contains less than three d-electrons, it will be labile.
- Or, if one or more e_g electrons are present, it will be labile.

Factors Affecting Lability of Complexes

No. of d electrons & electron configuration	Nature		Example
d^0	Labile		$[CaEDTA]^{2-}$
d^1; $t_{2g}^1 e_g^0$	Labile		$[Ti(H_2O)_6]^{3+}$
d^2; $t_{2g}^2 e_g^0$	Labile		$[V(phen)_3]^{3+}$
d^3; $t_{2g}^3 e_g^0$		Inert	$[V(H_2O)_6]^{3+}$
d^4 (high-spin); $t_{2g}^3 e_g^1$	Labile		$[Cr(H_2O)_6]^{3+}$
d^4 (low-spin); $t_{2g}^4 e_g^0$		Inert	$[Cr(CN)_6]^{4-}$
d^5 (high-spin); $t_{2g}^3 e_g^2$	Labile		$[Mn(H_2O)_6]^{2+}$
d^5 (low-spin); $t_{2g}^5 e_g^0$		Inert	$[Mn(CN)_6]^{4-}$
d^6 (high-spin); $t_{2g}^4 e_g^2$		Inert	$[Mn(H_2O)_6]^{2+}$
d^6 (low-spin); $t_{2g}^6 e_g^0$		Inert	$[Fe(CN)6]^{4-}$
d^7, d^8, d^9, d^{10}	Labile		

Properties of Ligand

✓ Size and charge of the ligand

- To some extent we can say that if the ligand is smaller in size and bearing higher charge it will form more stable complexes.
- For example usually F^- forms more stable complexes that Cl^-.
- In the case of neutral mono dentate ligands, high dipole moment and small size favour more stable complexes.

Properties of Ligand

- ✓ Basic character of ligands
- ✓ If the ligand is more basic then it will donate the electron pair more easily.
- ✓ So with increased basic character more stable complexes can be expected.
- ✓ Usually the ligands which bind strongly with H^+ form more stable complexes.
- ✓ This is observed for IA, IIA, 3d, 4f and 5f elements.

Properties of Ligand

✓Chelate effect

- This refers to the enhanced stability of a complex system containing one or more chelate rings as compared to the stability of a system which is as similar as possible without containing the ring. The chelate effect essentially is an entropy effect.
- $\Delta G^o = -RT\ln k$ and $\Delta G^o = \Delta H^o - T\Delta S^o$

ΔG is the change in the free energy of the system.

ΔH is the change in the enthalpy of the system.

ΔS is the change in the entropy (i.e. disorder) of the system.

T is the absolute temperature in Kelvin.

R is the universal gas constant, ($R = 8.314\ JK^{-1}\ mol^{-1}$ or $1.986\ calK^{-1}mol^{-1}$).

K is the equilibrium constant.

Properties of Ligand

✓Chelate effect

$\Delta G^o = -RT \ln k$

$\Delta G^o = \Delta H^o - T\Delta S^o$

- An increase in equilibrium constant may arise because ΔH^o becomes more negative, because ΔS^o becomes more positive or both.

- ΔH^o is contributed by the difference in energy of breaking M-OH_2 and forming M-L bond.

- However, the ΔH^o is very small and thus the chelate effect is entirely an entropy effect.

CHELATE EFFECT

$$[M(H_2O)_6] + 2NH_3 \rightleftharpoons [M(H_2O)_4(NH_3)_2] + 2H_2O$$

$$[M(H_2O)_6] + en \rightleftharpoons [M(H_2O)_4(en)] + 2H_2O$$

$$[M(H_2O)_6] + 6NH_3 \rightleftharpoons [M(NH_3)_6] + 6H_2O$$

$$[M(H_2O)_2] + 3en \rightleftharpoons [M(en)_3] + 6H_2O$$

- There is a limitation to the chelate effect when the chelating ligand is too long. It may pick on another metal ion in solution and this will effect no net change and will result in a decrease in ΔS^o. Most 6 and 7 membered rings are unstable.

Number of Chelate Rings

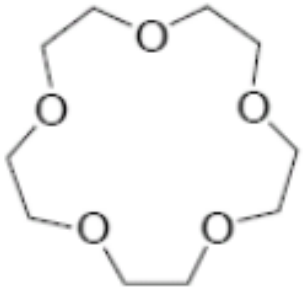
Metal complex	No. of rings	Values of log β						
		Mn (II)	Fe (II)	Co (II)	Ni (II)	Cu (II)	Zn (II)	Cd (II)
$M(NH_3)_4$	0	-	23.7	5.31	7.79	12.59	9.06	6.92
$M(en)_2$	2	4.9	7.7	10.9	14.5	20.2	11.2	10.3
M (trien)	3	4.9	7.8	11.0	14.1	20.5	12.1	10.0
M (tren)	3	2.8	8.8	12.8	14.0	18.8	14.6	12.3
$M(dien)_2$	4	7.0	10.4	14.1	18.9	21.3	14.4	13.8
M (penten)	5	9.4	11.2	15.8	19.3	22.4	16.2	16.2

Important Chelating Ligands

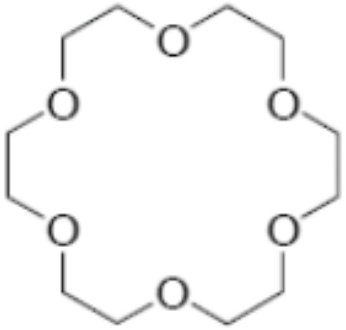
Macrocylic Ligands

Large ring compounds most commonly containing or O, N and/or S atoms

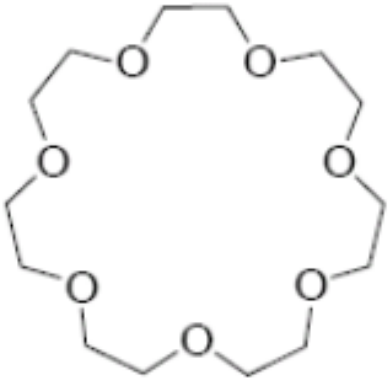
crown ether:



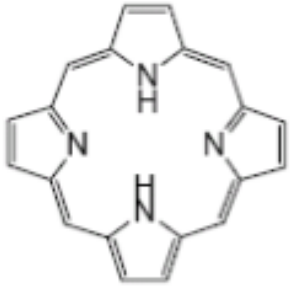
15-crown-5



18-crown-6

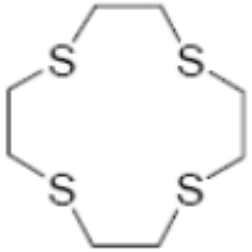


21-crown-7

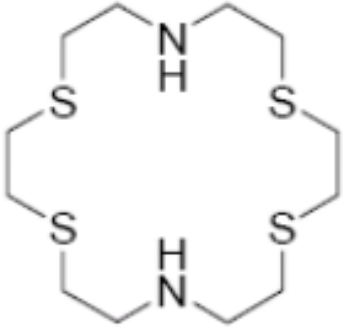


porphyrin

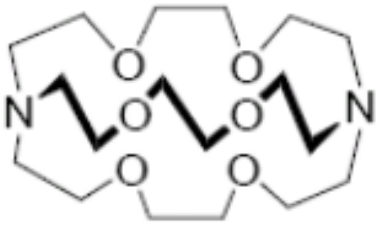
thia-crown ether:



aza-thia-crown:



cryptand:

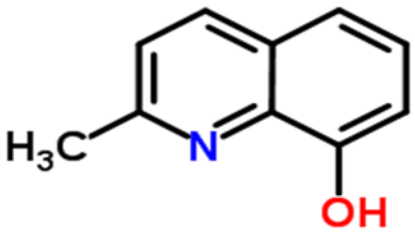


crypt-[222]

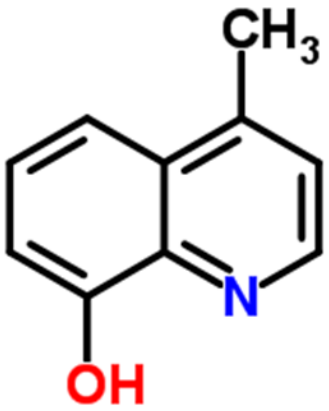
Properties of Ligand

✓Steric factors

- These are associated with the presence of bulky groups either attached to or near enough to a donor atom to cause mutual repulsion between the ligands and therefore, weakening of metal-ligands bonds.
- This, for instance, accounts for the lesser stability of metal complexes of 2-methyl-8-hydroxyquinoline than those of either 8-hydroxyquinoline itself or 4-methyl derivative.



Even when complex is formed, to get relieved from the steric hindrance the bond may dissociate. This reduces the stability of complex



Experimental determination of stability constant and composition of a complex

❑ Spectrophotometric method

➢ While formation of a complex a striking colour change also occurs.

- The absorption obeys Beer – Lambert's law:

$$A = \varepsilon . C . l$$

- A can be measured by using a spectrophotometer
- If ε and l are known then **C** can be calculated.
- Considering the following reaction,

$$M^{2+} + L \rightarrow ML^{2+}$$

$$K = [ML^{2+}] / [M^{2+}] [L]$$

It is known that, $C_M = [M^{2+}] + [ML^{2+}]$

$$C_L = [L] + [ML^{2+}]$$

$$A = \varepsilon_{[ML2+]} . C_{[ML2+]} . l$$

$$C_{[ML2+]} = A / \varepsilon_{[ML2+]} . l$$

Experimental determination of stability constant and composition of a complex

❑ Potentiometric method

➢ **Also known as Bjerrum method**

• **When ligand is a weak base or acid, there is competition between hydrogen ions and metal ions for the ligand.**

$$L + H^+ \rightarrow HL^+ \qquad K_a = [HL^+] / [L][H^+]$$
$$L + M^+ \rightarrow ML^+ \qquad K_F = [ML^+] / [L][M^+]$$

If C_H, C_M and C_L are the molar concentrations

$$C_H = [H^+] + [HL^+]$$
$$C_L = [L] + [ML^+] + [HL^+]$$
$$C_M = [M^+] + [ML^+]$$

Experimental determination of stability constant and composition of a complex

- Solving the equations by using the association constant of the ligand;

$$[ML^+] = C_L - C_H + [H^+] - (C_H - [H^+] / K_{a[H+]}$$

$$[M^+] = C_M - [ML^+]$$

$$[L] = C_H - [H^+] / K_{a[H+]}$$

- All the concentrations necessary for the formation constant can be determined if C_H, C_L, C_M, K_a and the $[H^+]$ are known.
- The $[H^+]$ can be determined potentiometrically using a pH meter.
- In order to get precise results the ligand must be a moderately weak base or acid.
- K_F value should be within 10^5 times of the **association constant**.

Exercise

1. A number of factors affect the stability of complexes. Some of these factors depend on the nature of the central ion, while others depend on the nature of the ligand. With reference to the properties of both central metal ion and the ligand, discuss the factors which affect the stability of complexes.

(a) Properties of the central metal ion
   (i) Charge and size of ion
   (ii) CFT and Irvin-Williams stability order
   (iii) Class a and class b metals –Ahrland and Chatt classification

(b) Properties of the ligand
   (i) size and charge of ligand
   (ii) Basic character
   (iii) Chelate effect
   (iv) Steric effect

ASSIGNMENT #1

1. Is there any relationship between the thermodynamic stability of a complex and its lability? Explain
2. The standard free energies of hydration of Cr^{3+} and Fe^{3+} ions are almost equal, yet $[Cr(H_2O)_6]^{3+}$ (d^3) undergoes substitution very slowly and $[Fe(H_2O)_6]^{3+}$ (d^3) very rapidly. Explain
3. Which of the following complexes do you expect to be labile and which is inert?

a. $[V(H_2O)_6]^{2+}$ b. $[Cr(H_2O)_6]^{3+}$ c. $[FeF_6]^{3-}$

d. $[Co(CN)_6]^{3-}$ e. $[CoF_6]^{3-}$ d. $[Ni(en)_3]^{2+}$

4. The successive stability constants for the species formed in aqueous solutions when ethylenediamine reacts with Ni^{2+} are as follows: log K_1 = 7.5, log K_2 = 6.4 and log K_3 = 4.4. Calculate β_3. What species are present in the equilibrium described by K_3? Give the geometry of the nickel containing species involved in the equilibrium described by K_3?

5. Describe three common methods to determine the stability (or formation) constants of complex compounds.